

PORT ISABEL HARBOR, TX, USA

WMO: 997369

Lat: 26.061N

Lon: 97.215W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.5	(c) 6.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 12.0	(k) 17.1	(l) 11.1	(m) 16.2	(n) 4.6	(o) 330	(p) 0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.8	(c) 31.8	(d) N/A	(e) 31.2	(f) N/A	(g) 30.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.4	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.3	(b) 10.4	(c) 9.3	(e) 3.1	(f) 34.0	(g) 2.3	(h) 0.9	(i) 1.5	(j) 34.6	(k) 0.1	(l) 35.2	(m) -1.1	(n) 35.7	(o) -2.8	(p) 36.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	16.0	17.4	20.2	23.5	26.0	28.2	28.5	29.0	27.6	25.6	20.8	18.2		
	DBStd	5.47	4.46	4.38	3.64	2.69	1.98	0.95	1.05	0.87	1.66	2.71	4.40	4.40		
	HDD10.0	19	10	4	1	0	0	0	0	0	0	0	2	2		
	HDD18.3	277	95	63	26	4	0	0	0	0	0	1	28	59		
	CDD10.0	4927	196	212	316	405	544	574	589	528	484	327	256			
	CDD18.3	2142	22	37	83	158	237	294	316	331	278	227	102	54		
	CDH23.3	20770	26	51	136	758	2085	3485	3861	4233	3321	2242	452	122		
	CDH26.7	6476	1	2	5	66	326	1156	1451	1777	1156	514	19	3		
(14) Wind	WSAvg	4.8	4.0	4.8	5.6	5.9	5.5	5.1	5.0	4.4	3.9	4.3	4.2	4.3		
(15) Precipitation	PrecAvg	638	25	30	34	40	50	59	48	47	135	90	45	34		
	PrecMax	857	90	105	185	194	152	197	226	143	316	266	161	124		
	PrecMin	364	0	1	2	0	0	8	2	0	25	0	0	2		
	PrecStd	168	22	28	40	46	40	46	59	37	79	72	40	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	27.0	27.1	30.2	30.5	31.9	32.5	32.9	32.6	31.4	28.5	26.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	23.9	24.9	25.5	28.3	29.5	30.8	31.5	31.8	31.3	30.3	27.3	25.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	22.7	23.7	24.7	27.3	28.8	30.2	30.8	31.2	30.6	29.6	26.5	24.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	21.6	22.5	24.0	26.4	28.3	29.9	30.3	30.8	30.2	28.9	25.7	23.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	5.8	4.8	4.2	3.4	3.4	3.7	3.8	4.3	4.9	5.5	6.2		
		MCDBR	7.0	6.5	4.6	5.1	3.7	3.9	4.0	4.3	4.4	4.7	5.1	6.1		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.372	0.384	0.402	0.453	0.470	0.438	0.457	0.442	0.444	0.397	0.381	0.375		
	taud		2.467	2.425	2.373	2.226	2.210	2.371	2.323	2.376	2.335	2.465	2.470	2.459		
	Ebn at Noon		878	893	894	852	832	854	837	851	842	868	861	856		
	Edh at Noon		98	109	121	144	146	123	129	122	124	104	97	95		
(44) All-Sky Solar Radiation	RadAvg		3.08	3.75	4.66	5.56	6.22	6.67	6.60	6.33	5.26	4.61	3.52	2.80		
	RadStd		0.34	0.54	0.47	0.47	0.36	0.24	0.45	0.26	0.36	0.41	0.36	0.37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118

Nomenclature: See separate page